



# **LIBRARY MANAGEMENT SYSTEM**

**Project Submitted By**

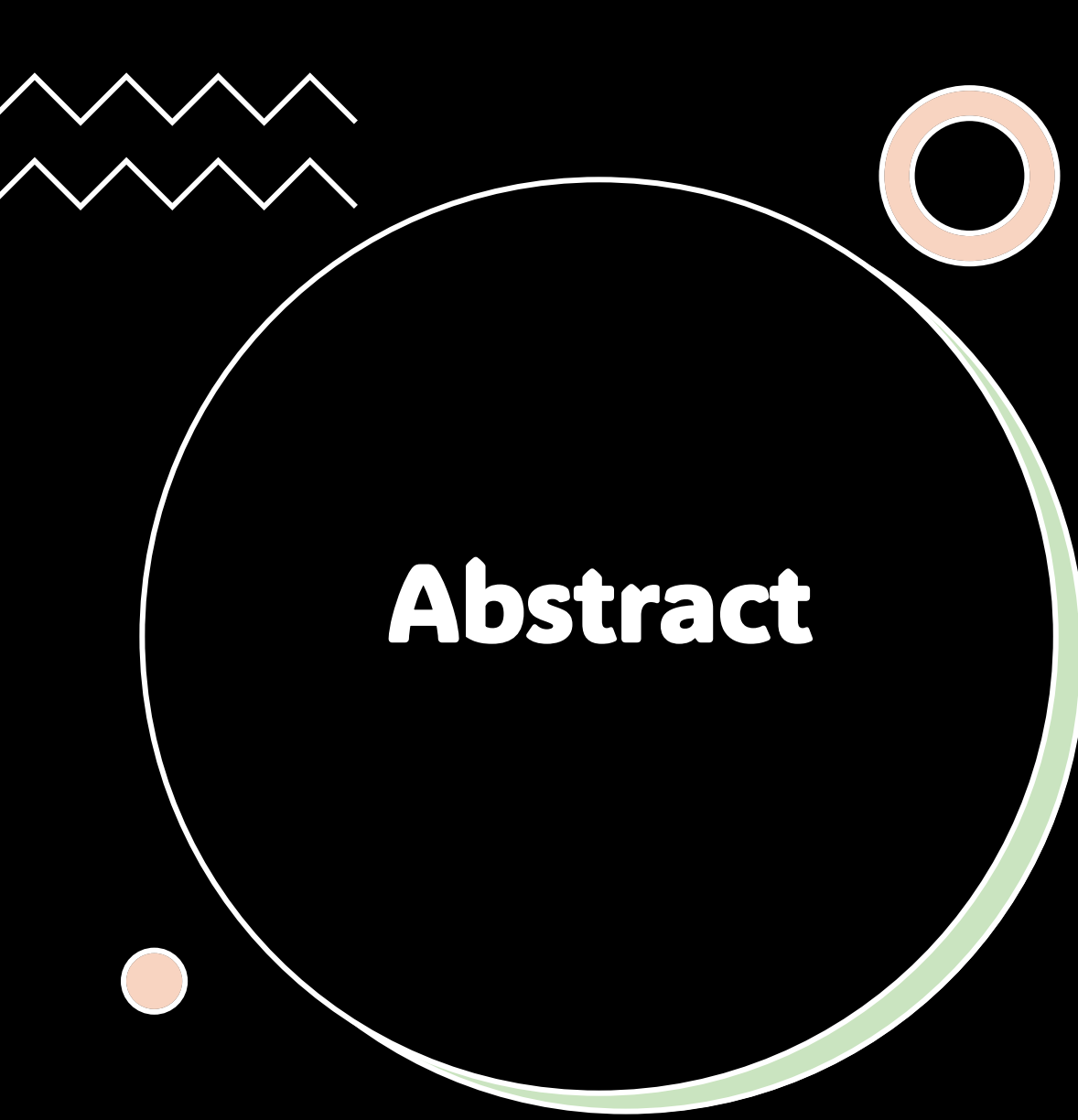
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**Year: III Year B.Tech**

**College: SRM University**

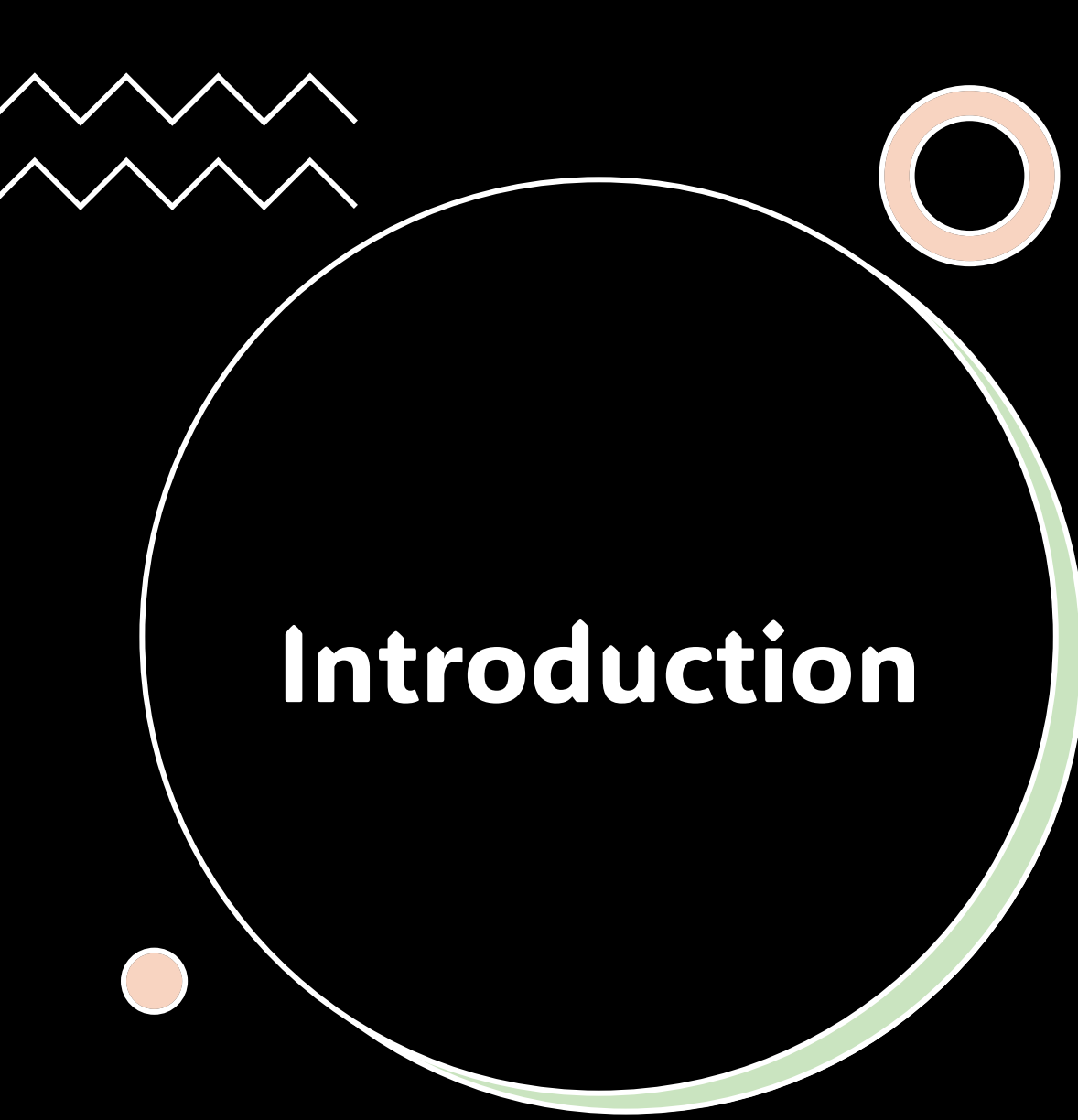




# **Abstract**

The Library Management System is a database project developed using MySQL. The system is designed to manage library operations such as storing book information, managing member details, issuing books, and tracking returned books. This project demonstrates the use of relational databases, primary keys, foreign keys, and SQL queries.





# Introduction

Libraries maintain a large number of books and members. Managing these records manually is difficult and time-consuming.

The Library Management System provides a digital solution for storing and managing library data efficiently.

- The system allows:
  - Storing book details
  - Managing member information
  - Issuing books
  - Returning books
  - Tracking library records
- 

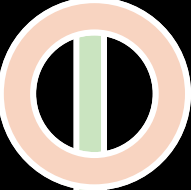


# Objectives

## The objectives of this project are:

- To create a database for managing library records.
  - To store book information.
  - To maintain member details.
  - To manage book issue records.
  - To manage book return records.
  - To practice SQL commands and database concepts.
- 

# Software Requirements



| <u>Software</u> | <u>Purpose</u>       |
|-----------------|----------------------|
| MySQL Server    | Database Management  |
| MySQL Workbench | SQL Development Tool |
| Windows 11      | Operating System     |





# Database Design

## Database Name

- CREATE DATABASE library\_management;

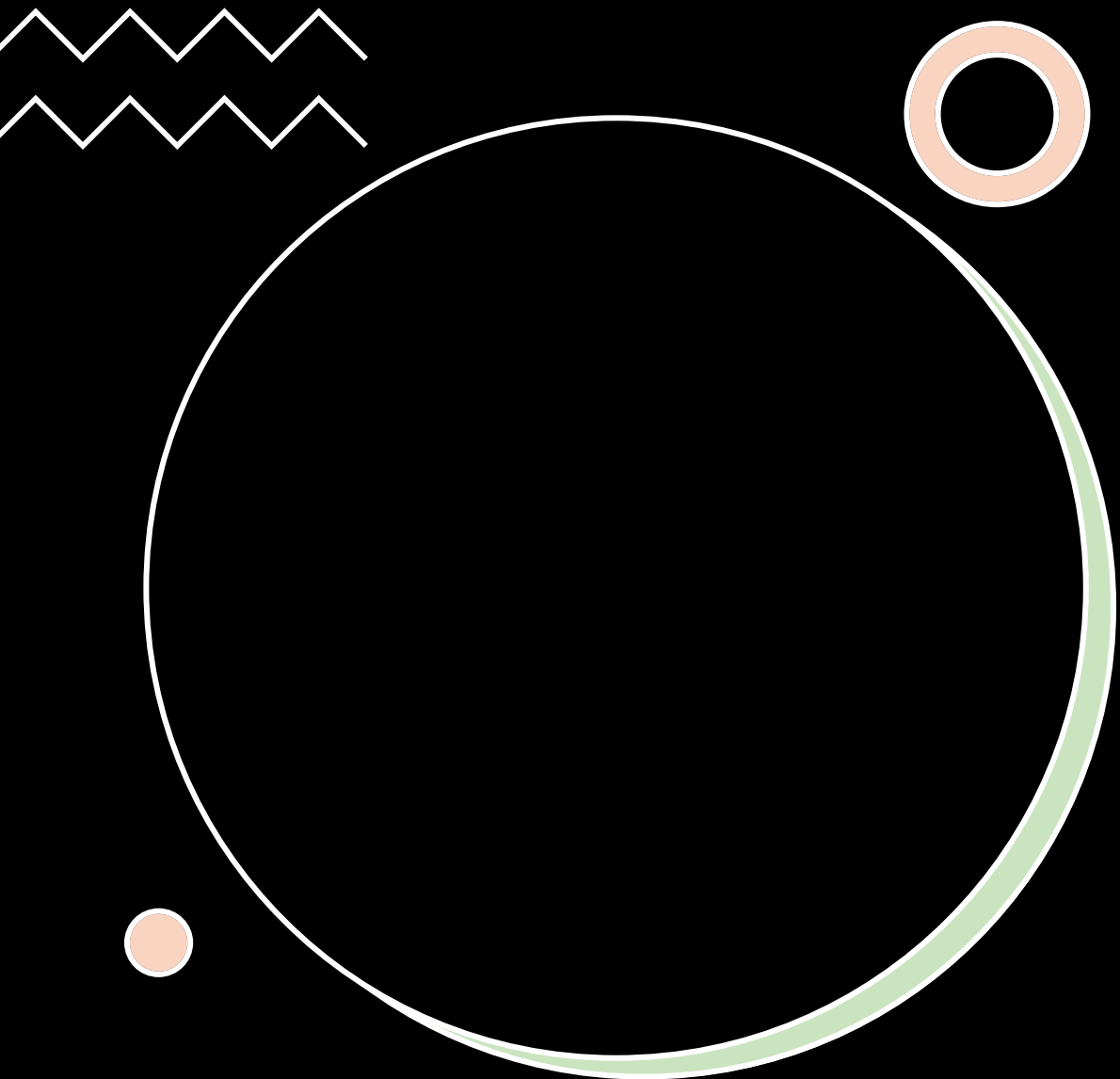
## Select Database

- USE library\_management;

## Table Creation

### 1. Books Table

- CREATE TABLE books (  
    book\_id INT PRIMARY KEY,  
    title VARCHAR(100),  
    author VARCHAR(100),  
    quantity INT  
);
  - **Purpose**
  - Stores information about books available in the library.
- 



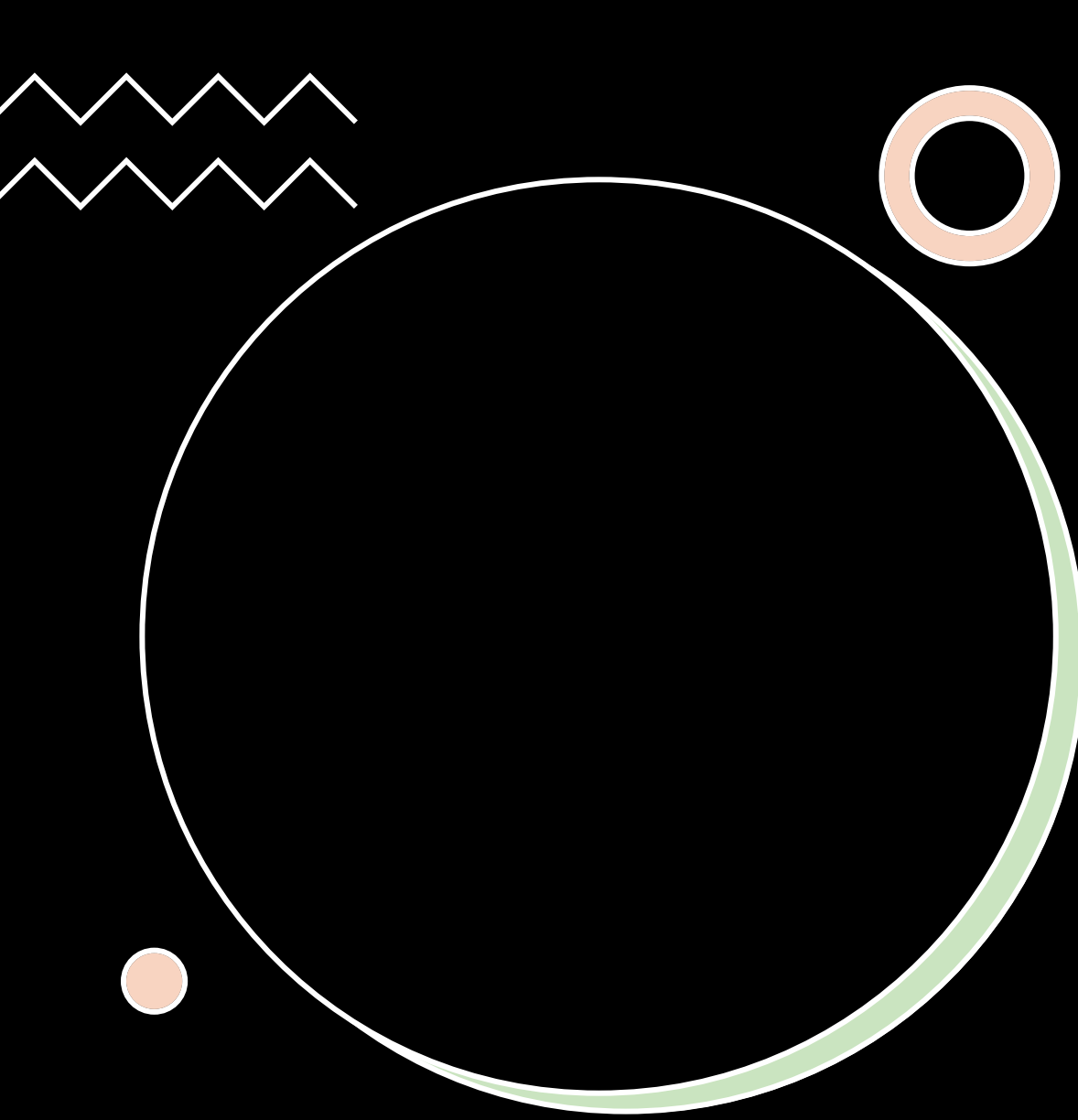
## 2. Members Table

- CREATE TABLE members (  
    member\_id INT PRIMARY KEY,  
    member\_name VARCHAR(100),  
    phone VARCHAR(15),  
    email VARCHAR(100)  
);
- **Purpose**
- Stores member information.

## 3. Issue Books Table

- CREATE TABLE issue\_books (  
    issue\_id INT PRIMARY KEY,  
    book\_id INT,  
    member\_id INT,  
    issue\_date DATE,  
    FOREIGN KEY (book\_id) REFERENCES books(book\_id),  
    FOREIGN KEY (member\_id) REFERENCES members(member\_id)  
);
- **Purpose**
- Stores information about issued books.

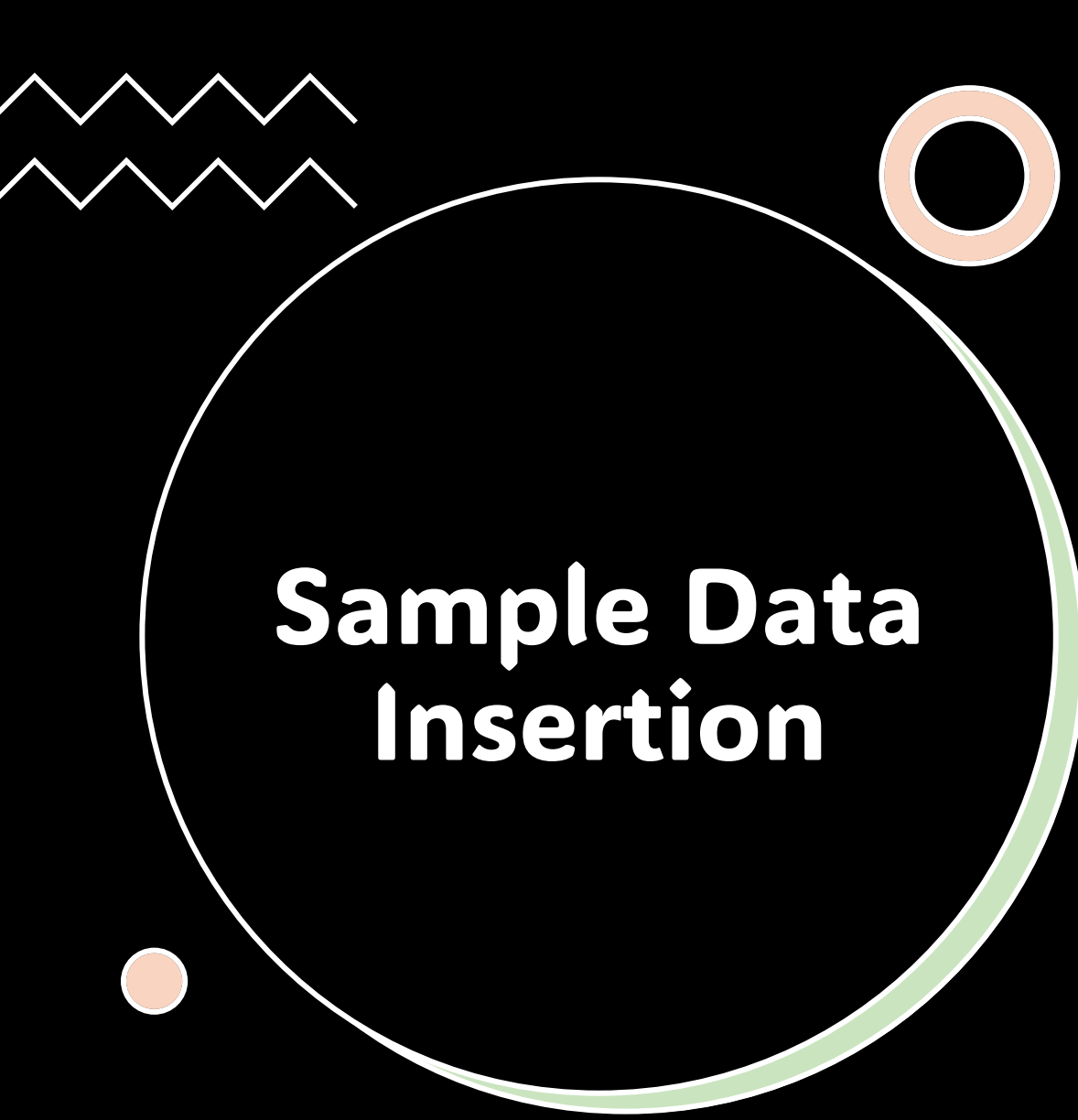




#### 4. Return Books Table

- ```
CREATE TABLE return_books (  
    return_id INT PRIMARY KEY,  
    issue_id INT,  
    return_date DATE,  
    FOREIGN KEY (issue_id) REFERENCES  
    issue_books(issue_id)  
);
```
  - **Purpose**
  - Stores information about returned books.
- 





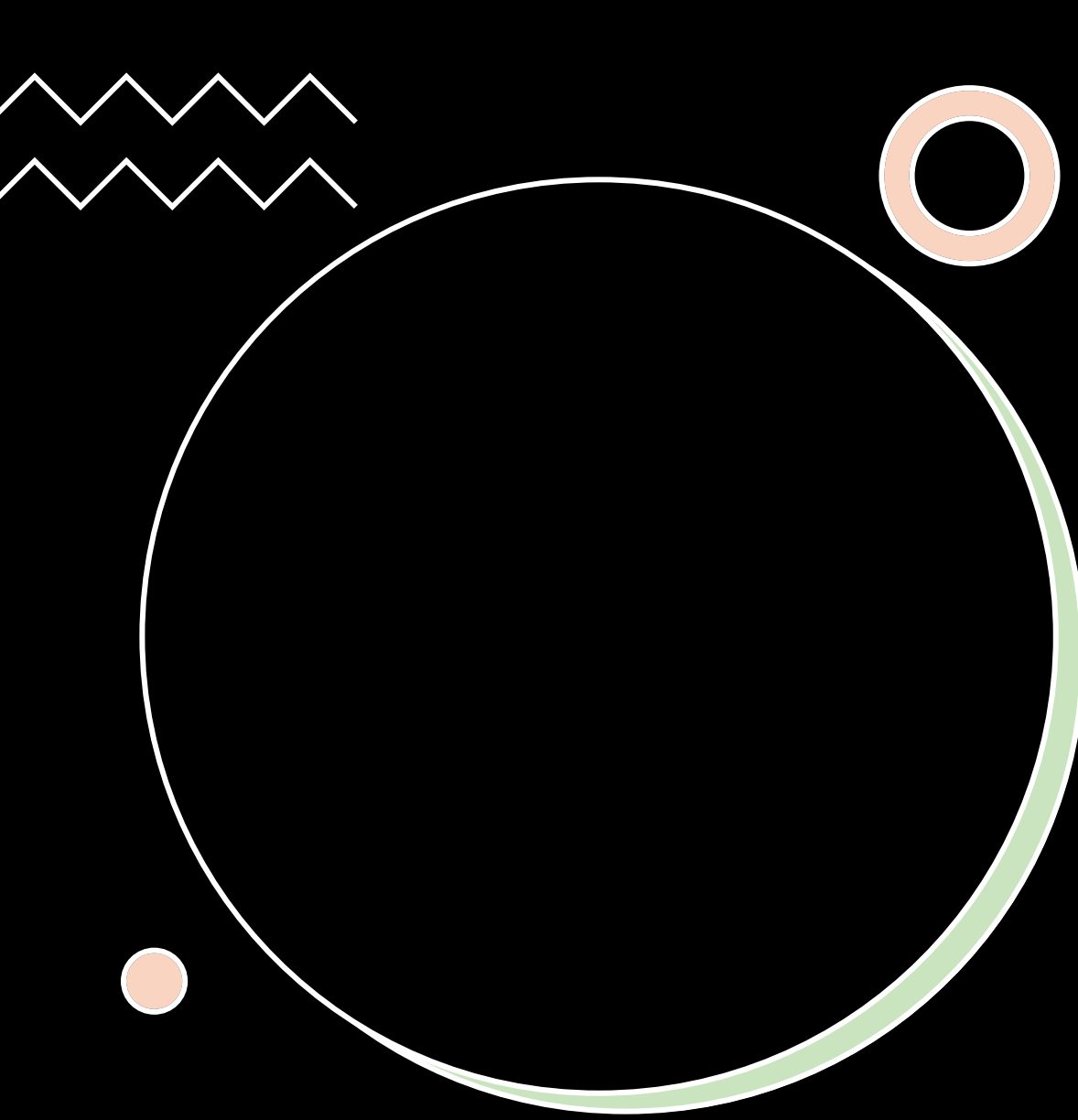
# Sample Data Insertion

## Insert Books

- INSERT INTO books VALUES  
(1,'Database Systems','Korth',10),  
(2,'Python Programming','John Zelle',15),  
(3,'Java Fundamentals','Herbert Schildt',8);

## Insert Members

- INSERT INTO members VALUES  
(101,'Saran Kumar','9876543210','saran@gmail.com'),  
(102,'Rahul','9876543211','rahul@gmail.com'),  
(103,'Priya','9876543212','priya@gmail.com');
- 

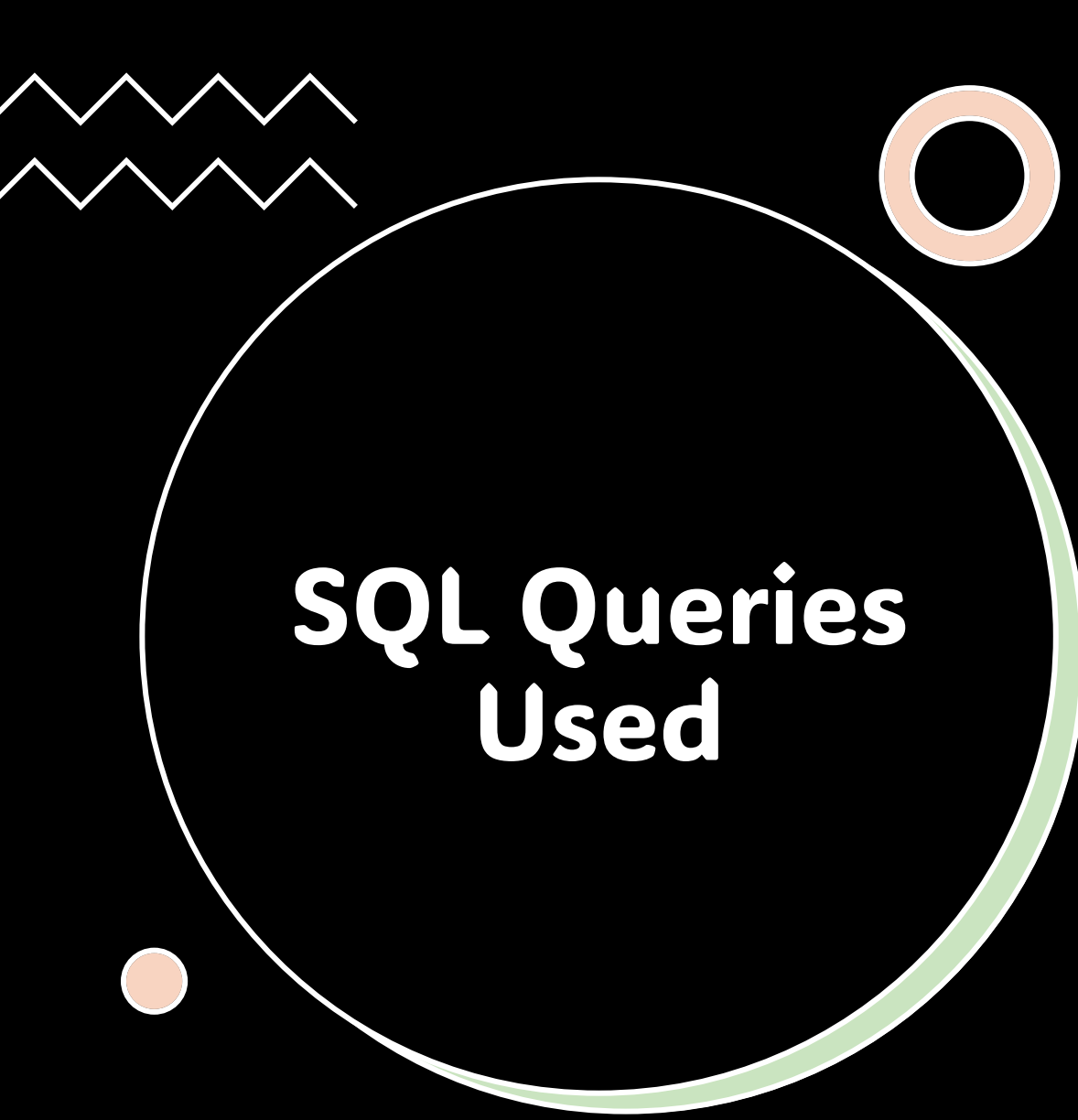


### Insert Issue Records

- INSERT INTO issue\_books VALUES  
(1,1,101,'2026-06-12'),  
(2,2,102,'2026-06-13');

### Insert Return Records

- INSERT INTO return\_books VALUES  
(1,1,'2026-06-20');
- 



# SQL Queries Used

- **Display All Books**

```
SELECT * FROM books;
```

- **Display All Members**

```
SELECT * FROM members;
```

- **Display Issued Books**

```
SELECT * FROM issue_books;
```

- **Display Returned Books**

```
SELECT * FROM return_books;
```

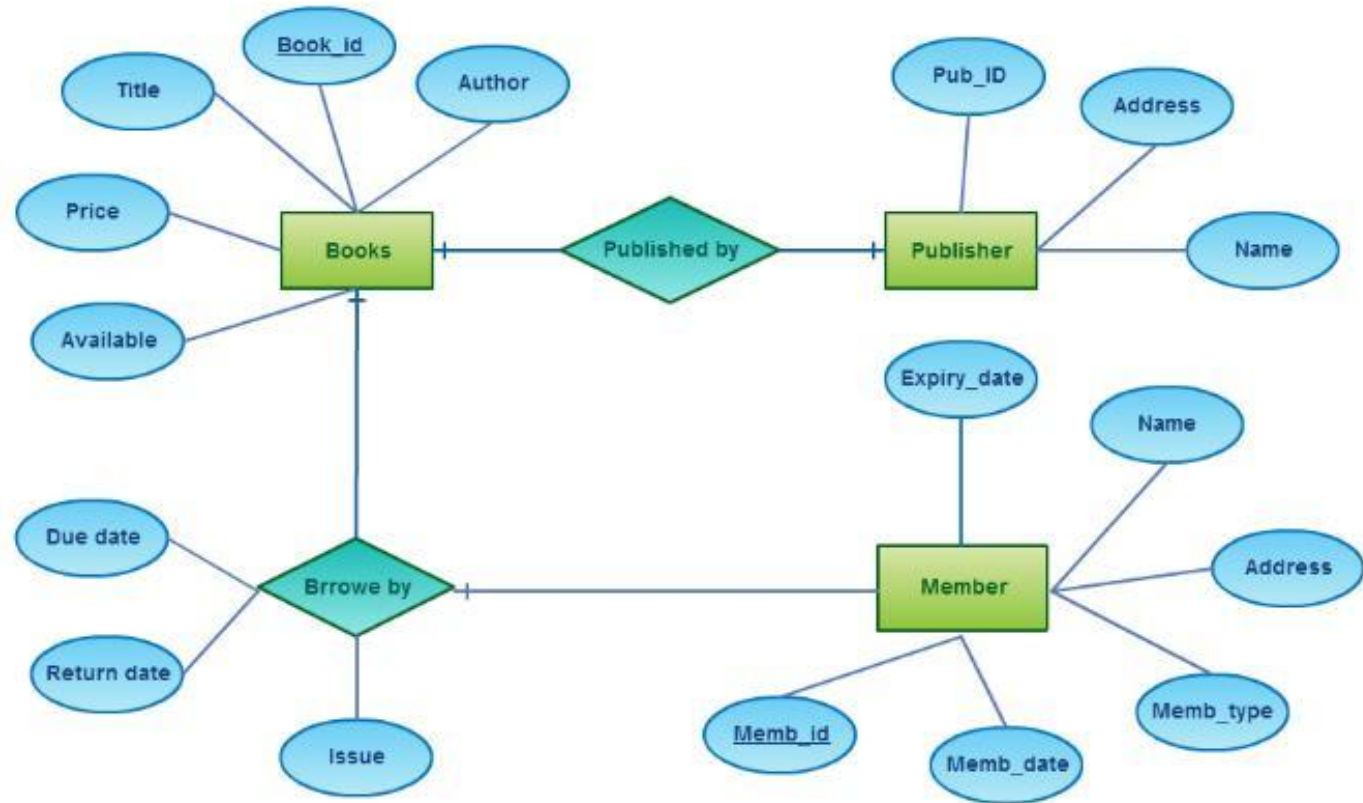
- **Show All Tables**

```
SHOW TABLES;
```



# ER Diagram

E-R Diagram of Library Management System



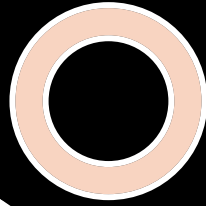


# Results

The Library Management System database was successfully created using MySQL.

The following operations were performed successfully:

- Database Creation
  - Table Creation
  - Data Insertion
  - Data Retrieval
  - Foreign Key Relationships
  - SQL Query Execution
- 



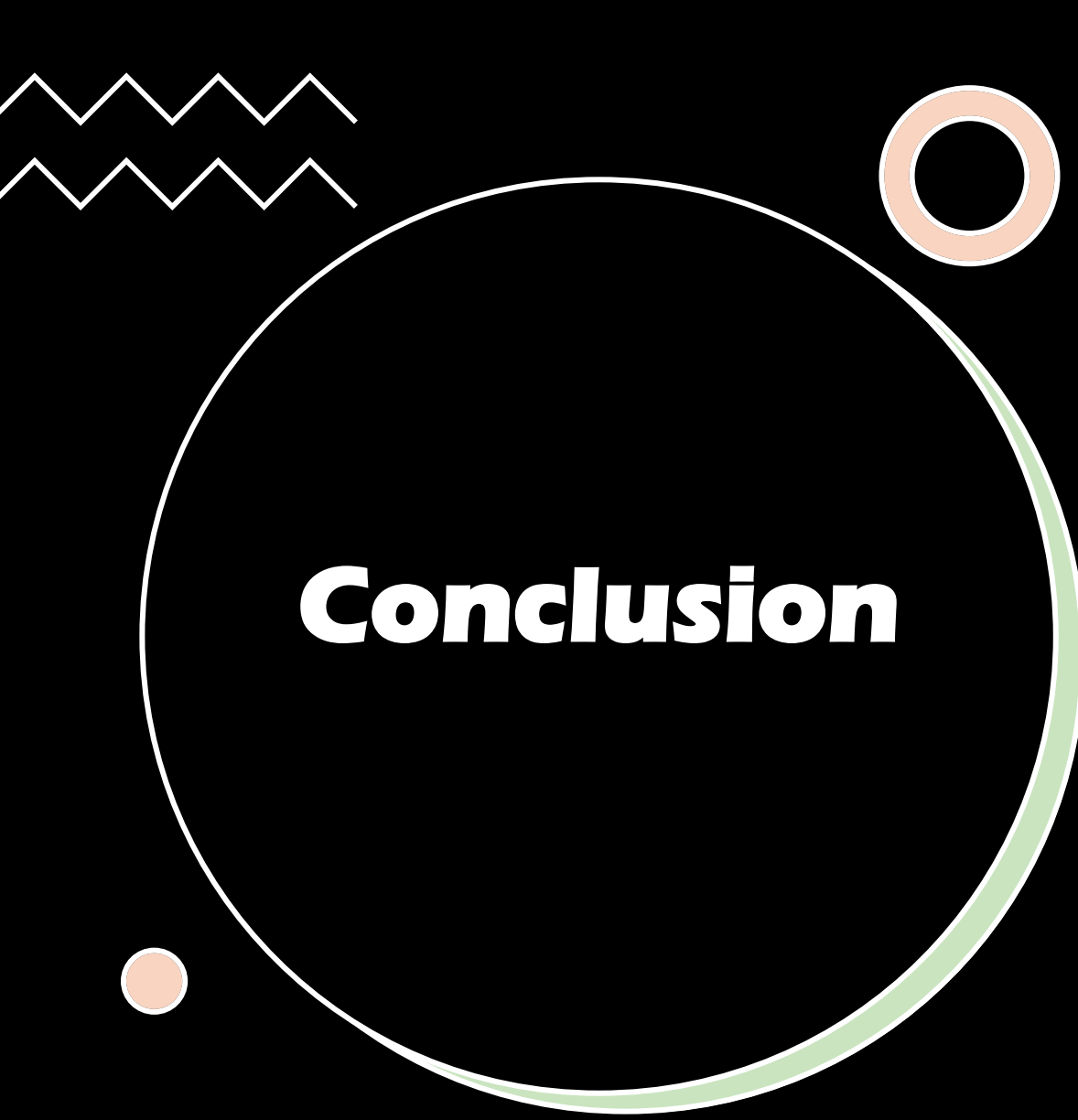
## Advantages

- Easy to manage library records.
- Reduces manual work.
- Fast data retrieval.
- Maintains data consistency.
- Demonstrates relational database concepts.

## Future Enhancements

- Fine Calculation
- Login System
- Book Search Functionality
- Report Generation
- Web-Based Interface





# **Conclusion**

The Library Management System was successfully implemented using MySQL. The project demonstrates the use of SQL commands, relational database design, primary keys, foreign keys, and data management techniques. The system efficiently manages books, members, issued books, and returned books.





Navigator

## SCHEMAS

Filter objects

- eg
- library\_management
  - Tables
    - books
    - issue\_books
    - members
    - return\_books
  - Views
  - Stored Procedures
  - Functions
- sys

Administration Schemas

Information

## Connection Details

local host  
127.0.0.1  
3306  
root  
Use  
root@localhost  
Use  
TLS\_AES\_128\_GCM\_SHA256  
Server  
MySQL Community

Object Info Session

library\_management x Administration - Server Status



Limit to 1000 rows

```
36 • INSERT INTO issue_books VALUES
37 (1, 1, 101, '2026-06-12'),
38 (2, 2, 102, '2026-06-13');
39 • INSERT INTO return_books VALUES
40 (1, 1, '2026-06-20');
41 • USE library_management;
42 • SHOW TABLES;
```

Result Grid Filter Rows: Export: Wrap Cell Content:

|                              |
|------------------------------|
| Tables_in_library_management |
| books                        |
| issue_books                  |
| members                      |
| return_books                 |

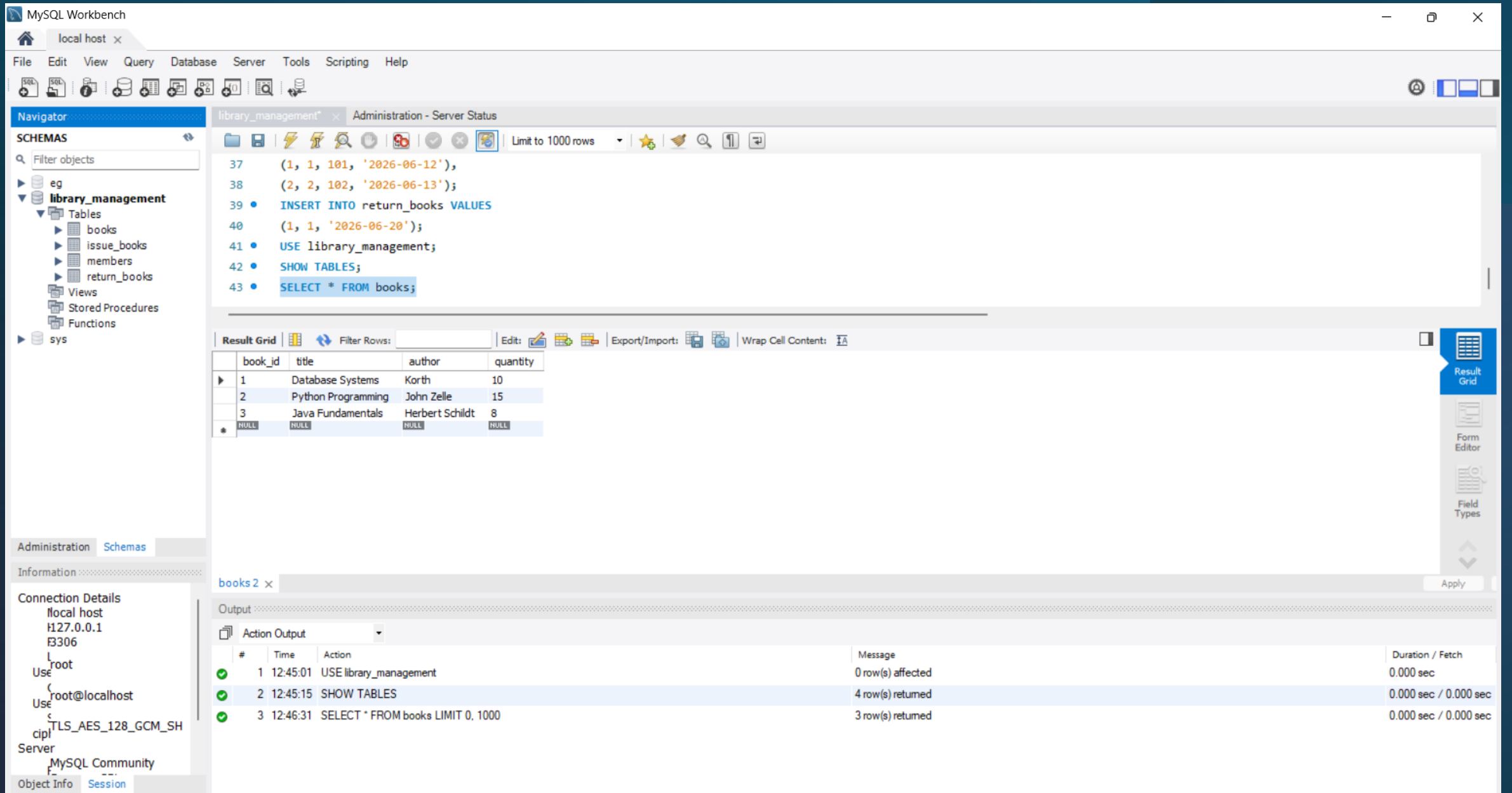
Result 1 x

Output

Action Output

| #   | Time     | Action                 | Message           | Du   |
|-----|----------|------------------------|-------------------|------|
| ✓ 1 | 12:45:01 | USE library_management | 0 row(s) affected | 0.00 |
| ✓ 2 | 12:45:15 | SHOW TABLES            | 4 row(s) returned | 0.00 |





MySQL Workbench

local host x

File Edit View Query Database Server Tools Scripting Help

Navigator: library\_management\* x Administration - Server Status

**SCHEMAS**

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Administration Schemas

Information

Connection Details

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127.0.0.1  
3306  
root  
Use root@localhost  
Use TLS\_AES\_128\_GCM\_SHA256  
Server MySQL Community

Object Info Session

38 (2, 2, 102, '2026-06-13');  
39 • INSERT INTO return\_books VALUES  
40 (1, 1, '2026-06-20');  
41 • USE library\_management;  
42 • SHOW TABLES;  
43 • SELECT \* FROM books;  
44 • SELECT \* FROM members;

Limit to 1000 rows

Result Grid

|   | member_id | member_name | phone      | email           |
|---|-----------|-------------|------------|-----------------|
| ▶ | 101       | Saran Kumar | 9876543210 | saran@gmail.com |
|   | 102       | Rahul       | 9876543211 | rahul@gmail.com |
|   | 103       | Priya       | 9876543212 | priya@gmail.com |
| * | NULL      | NULL        | NULL       | NULL            |

members 3 x

Output

Action Output

| #   | Time     | Action                              | Message           | Duration / Fetch      |
|-----|----------|-------------------------------------|-------------------|-----------------------|
| ✓ 1 | 12:45:01 | USE library_management              | 0 row(s) affected | 0.000 sec             |
| ✓ 2 | 12:45:15 | SHOW TABLES                         | 4 row(s) returned | 0.000 sec / 0.000 sec |
| ✓ 3 | 12:46:31 | SELECT * FROM books LIMIT 0, 1000   | 3 row(s) returned | 0.000 sec / 0.000 sec |
| ✓ 4 | 12:47:15 | SELECT * FROM members LIMIT 0, 1000 | 3 row(s) returned | 0.000 sec / 0.000 sec |

MySQL Workbench

local host x

File Edit View Query Database Server Tools Scripting Help

Navigator: library\_management\* x Administration - Server Status

Limit to 1000 rows

**SCHEMAS**

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Administration Schemas

Information

Connection Details

local host  
127.0.0.1  
3306  
root  
Use root@localhost  
Use TLS\_AES\_128\_GCM\_SHA256  
Server MySQL Community

Object Info Session

```
39 • INSERT INTO return_books VALUES
40 (1, 1, '2026-06-20');
41 • USE library_management;
42 • SHOW TABLES;
43 • SELECT * FROM books;
44 • SELECT * FROM members;
45 • SELECT * FROM issue_books;
```

Result Grid

|   | issue_id | book_id | member_id | issue_date |
|---|----------|---------|-----------|------------|
| ▶ | 1        | 1       | 101       | 2026-06-12 |
|   | 2        | 2       | 102       | 2026-06-13 |
| * | NULL     | NULL    | NULL      | NULL       |

Filter Rows: Edit: Export/Import: Wrap Cell Content: Apply

issue\_books 4 x

Output

Action Output

| #   | Time     | Action                                  | Message           | Duration / Fetch      |
|-----|----------|-----------------------------------------|-------------------|-----------------------|
| ✓ 1 | 12:45:01 | USE library_management                  | 0 row(s) affected | 0.000 sec             |
| ✓ 2 | 12:45:15 | SHOW TABLES                             | 4 row(s) returned | 0.000 sec / 0.000 sec |
| ✓ 3 | 12:46:31 | SELECT * FROM books LIMIT 0, 1000       | 3 row(s) returned | 0.000 sec / 0.000 sec |
| ✓ 4 | 12:47:15 | SELECT * FROM members LIMIT 0, 1000     | 3 row(s) returned | 0.000 sec / 0.000 sec |
| ✓ 5 | 12:47:47 | SELECT * FROM issue_books LIMIT 0, 1000 | 2 row(s) returned | 0.000 sec / 0.000 sec |

MySQL Workbench

local host x

File Edit View Query Database Server Tools Scripting Help

Navigator: library\_management\* x Administration - Server Status

Limit to 1000 rows

**SCHEMAS**

Filter objects

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- sys

Administration Schemas

Information

Connection Details

localhost  
127.0.0.1  
3306  
root  
Use root@localhost  
Use root@localhost  
cpi TLS\_AES\_128\_GCM\_SHA256  
Server MySQL Community

Object Info Session

```
40 (1, 1, '2026-06-20');
41 USE library_management;
42 SHOW TABLES;
43 SELECT * FROM books;
44 SELECT * FROM members;
45 SELECT * FROM issue_books;
46 SELECT * FROM return_books;
```

Result Grid

|   | return_id | issue_id | return_date |
|---|-----------|----------|-------------|
| ▶ | 1         | 1        | 2026-06-20  |
| * | NULL      | NULL     | NULL        |

return\_books 5 x

Output

Action Output

| #   | Time     | Action                                   | Message           | Duration / Fetch      |
|-----|----------|------------------------------------------|-------------------|-----------------------|
| ✓ 1 | 12:45:01 | USE library_management                   | 0 row(s) affected | 0.000 sec             |
| ✓ 2 | 12:45:15 | SHOW TABLES                              | 4 row(s) returned | 0.000 sec / 0.000 sec |
| ✓ 3 | 12:46:31 | SELECT * FROM books LIMIT 0, 1000        | 3 row(s) returned | 0.000 sec / 0.000 sec |
| ✓ 4 | 12:47:15 | SELECT * FROM members LIMIT 0, 1000      | 3 row(s) returned | 0.000 sec / 0.000 sec |
| ✓ 5 | 12:47:47 | SELECT * FROM issue_books LIMIT 0, 1000  | 2 row(s) returned | 0.000 sec / 0.000 sec |
| ✓ 6 | 12:49:32 | SELECT * FROM return_books LIMIT 0, 1000 | 1 row(s) returned | 0.000 sec / 0.000 sec |

MySQL Workbench

local host x

File Edit View Query Database Server Tools Scripting Help

Navigator: library\_management\* x Administration - Server Status

Limit to 1000 rows

SCHEMAS

Filter objects

eg

library\_management

- Tables
  - books
  - issue\_books
  - members
  - return\_books
- Views
- Stored Procedures
- Functions

sys

Administration Schemas

Information

Connection Details

local host  
127.0.0.1  
3306  
root  
Use root@localhost  
Use TLS\_AES\_128\_GCM\_SHA256  
Server MySQL Community

Object Info Session

```
41 • USE library_management;
42 • SHOW TABLES;
43 • SELECT * FROM books;
44 • SELECT * FROM members;
45 • SELECT * FROM issue_books;
46 • SELECT * FROM return_books;
47 • DESC books;
```

Result Grid

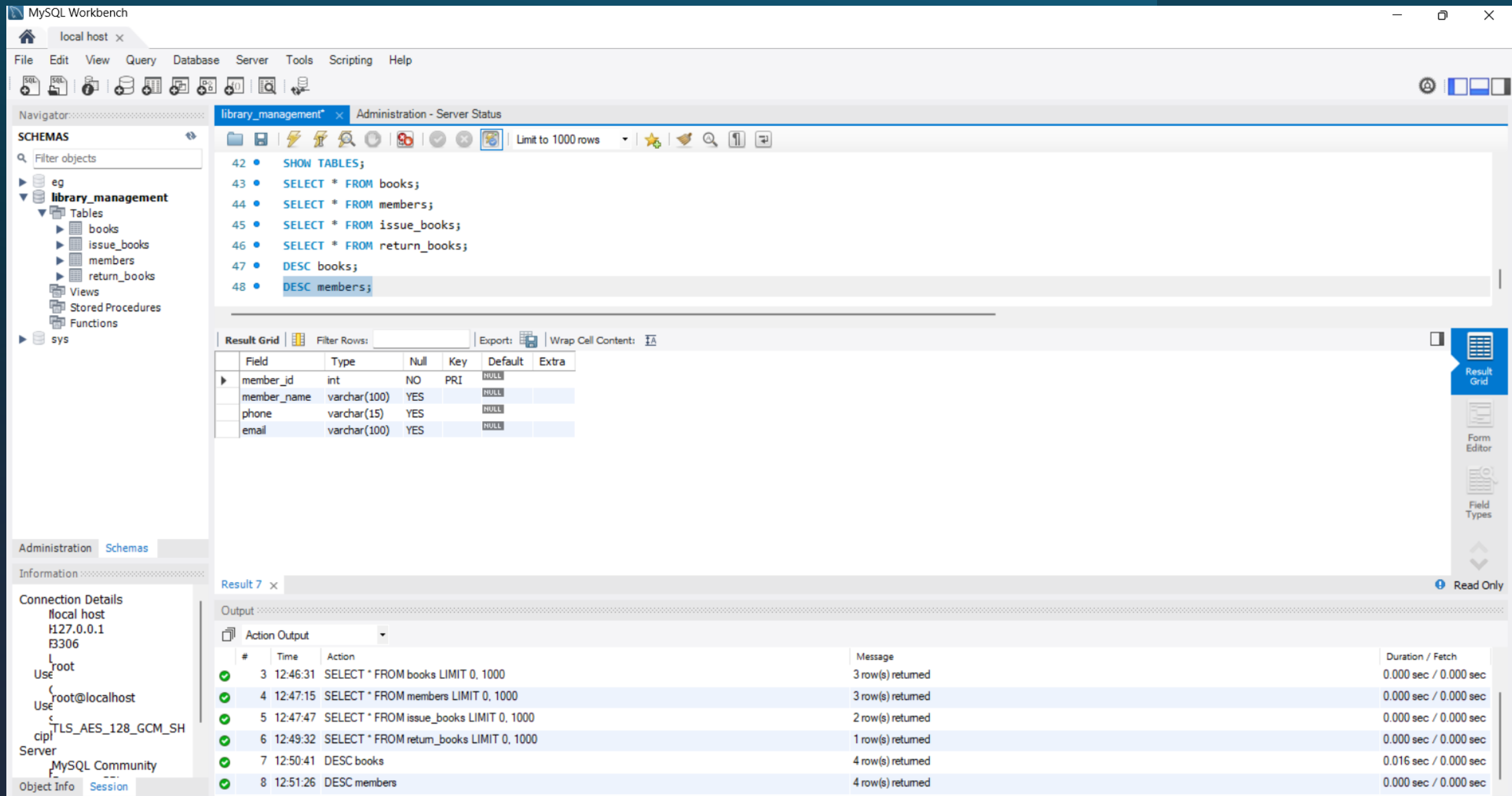
|   | Field    | Type         | Null | Key | Default | Extra |
|---|----------|--------------|------|-----|---------|-------|
| ▶ | book_id  | int          | NO   | PRI | NULL    |       |
|   | title    | varchar(100) | YES  |     | NULL    |       |
|   | author   | varchar(100) | YES  |     | NULL    |       |
|   | quantity | int          | YES  |     | NULL    |       |

Result 6 x

Output

Action Output

| #   | Time     | Action                                   | Message           | Duration / Fetch      |
|-----|----------|------------------------------------------|-------------------|-----------------------|
| ✓ 2 | 12:45:15 | SHOW TABLES                              | 4 row(s) returned | 0.000 sec / 0.000 sec |
| ✓ 3 | 12:46:31 | SELECT * FROM books LIMIT 0, 1000        | 3 row(s) returned | 0.000 sec / 0.000 sec |
| ✓ 4 | 12:47:15 | SELECT * FROM members LIMIT 0, 1000      | 3 row(s) returned | 0.000 sec / 0.000 sec |
| ✓ 5 | 12:47:47 | SELECT * FROM issue_books LIMIT 0, 1000  | 2 row(s) returned | 0.000 sec / 0.000 sec |
| ✓ 6 | 12:49:32 | SELECT * FROM return_books LIMIT 0, 1000 | 1 row(s) returned | 0.000 sec / 0.000 sec |
| ✓ 7 | 12:50:41 | DESC books                               | 4 row(s) returned | 0.016 sec / 0.000 sec |



MySQL Workbench

local host x

File Edit View Query Database Server Tools Scripting Help

Navigator

library\_management\* x Administration - Server Status

SCHEMAS

Filter objects

eg

library\_management

Tables

books

issue\_books

members

return\_books

Views

Stored Procedures

Functions

sys

43 • SELECT \* FROM books;

44 • SELECT \* FROM members;

45 • SELECT \* FROM issue\_books;

46 • SELECT \* FROM return\_books;

47 • DESC books;

48 • DESC members;

49 • SELECT m.member\_name,

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

|   | member_name | title              | issue_date |
|---|-------------|--------------------|------------|
| ▶ | Saran Kumar | Database Systems   | 2026-06-12 |
|   | Rahul       | Python Programming | 2026-06-13 |

Administration Schemas

Information

Connection Details

local host

127.0.0.1

3306

root

Use

root@localhost

Use

TLS\_AES\_128\_GCM\_SHA256

Server

MySQL Community

Object Info Session

Result 8 x

Read Only

Output

Action Output

| #   | Time     | Action                                                                                          | Message           | Duration / Fetch      |
|-----|----------|-------------------------------------------------------------------------------------------------|-------------------|-----------------------|
| ✓ 4 | 12:47:15 | SELECT * FROM members LIMIT 0, 1000                                                             | 3 row(s) returned | 0.000 sec / 0.000 sec |
| ✓ 5 | 12:47:47 | SELECT * FROM issue_books LIMIT 0, 1000                                                         | 2 row(s) returned | 0.000 sec / 0.000 sec |
| ✓ 6 | 12:49:32 | SELECT * FROM return_books LIMIT 0, 1000                                                        | 1 row(s) returned | 0.000 sec / 0.000 sec |
| ✓ 7 | 12:50:41 | DESC books                                                                                      | 4 row(s) returned | 0.016 sec / 0.000 sec |
| ✓ 8 | 12:51:26 | DESC members                                                                                    | 4 row(s) returned | 0.000 sec / 0.000 sec |
| ✓ 9 | 12:52:20 | SELECT m.member_name, b.title, i.issue_date FROM issue_books i JOIN members m ON i.member_id... | 2 row(s) returned | 0.000 sec / 0.000 sec |